VR World Director: Camera-Path Authoring Inside a World Model

Mixed Reality Project Proposal
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I. DESCRIPTION OF THE PROJECT

We will build a VR editor in which a user directs a virtual camera *inside* a world generated by a 3D world model, and test whether VR makes shot composition easier than a desktop editor. Our backbone is Puffin-World [1], an open-source unified multimodal world model (code, weights and data released) that, given one initial view and a camera trajectory, jointly generates RGB and depth for the remaining views and fuses them into a 3D point cloud. Its 9-channel *Omni-Camera* condition combines a gravity-aware absolute field (up-vector, latitude [3]) with a relative ray map, and *physics propagation* carries the gravity direction perceived in the reference view along the whole trajectory, which keeps horizons stable under the rolls, tilts and pans that cinematographers use. **Input:** a seed image or text prompt, and a camera path authored with Meta Quest controllers. **Output:** the generated RGB-D video along that path, the exported trajectory (poses + intrinsics), and the growing point cloud of the world. While new views are generated asynchronously on a GPU server, the user keeps previewing and editing against the geometry that already exists. Atlas [2], World Labs' camera-controlled omni world model, is a stretch goal if research access is granted.

II. WORK PACKAGES AND TIMELINE

Platform. PC-VR: Unity 6 with OpenXR on a Meta Quest 3 via Quest Link (C#); Python/PyTorch inference server (Puffin-World reference stack: PyTorch 2.7, CUDA 12.6) on a remote GPU, linked by WebSocket. Owners below are initial; everyone contributes to integration and the study.

WP0 Literature review and baseline setup (all, Oct 5–18). Run Puffin-World's world-modeling, physics-perception and self-calibration demos; measure memory and time per chunk for the Base (SD3.5-Medium, 2.5B) and Pro (SD3.5-Large, 8.1B) checkpoints. *Outcome:* reproducible inference and a latency budget for the interaction design.

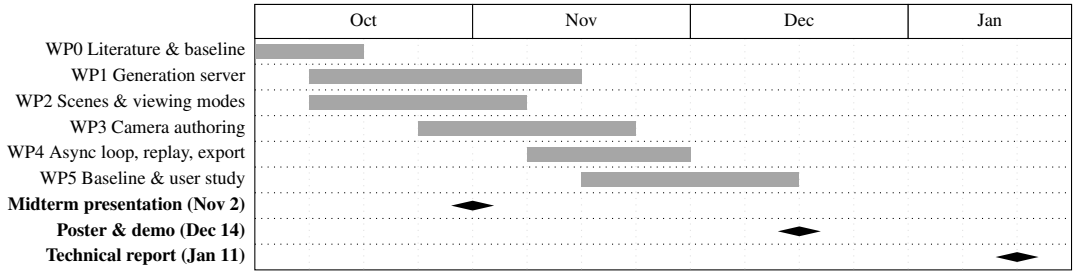
WP1 Generation server (Darius, Oct 12–Nov 15). Wrap trajectory→RGB-D→point-cloud generation behind a job queue; long paths are split into chunks of up to 7 target views, chained through Puffin-World's latent carry-over; results stream back per chunk. *Outcome:* API taking a seed and a trajectory and returning frames, depth and a fused cloud.

WP2 Scene import and viewing modes (Rayane, Oct 12–Nov 8). Seed three scenes (indoor, outdoor, one text-to-3D) by generating an exploratory 360° pan plus a forward walk, then load the fused clouds in Unity with a tabletop *miniature* (world-in-miniature [4]) and a 1:1 *immersive* mode. *Outcome:* three navigable scenes in VR.

WP3 Camera authoring (Baudoin, Oct 26–Nov 22). Place, move and delete keyframes with the controllers; a handheld virtual viewfinder shows the framing rendered from the point cloud; Catmull-Rom interpolation for position, quaternion SLERP for orientation, per-keyframe FoV, and a timeline for timing. *Outcome:* editable paths in both modes.

WP4 Asynchronous loop, replay and export (Darius, Baudoin, Nov 9–29). Send edited path segments for generation, fuse returned geometry into the live scene, replay the video on an in-VR screen or along the path, export MP4 and the trajectory (RealEstate10K-style [5] text file and JSON).

WP5 Desktop baseline and user study (Rayane, Nov 16–Dec 10). Mouse-and-keyboard editor in Unity sharing the same backend; run and analyse the study (Sec. III).



Challenges, planned solutions and alternatives.

Challenge	Planned solution	Alternative if it fails
C1 Latency: 50 diffusion steps over 8 views at 640×640 per pass [1], far from interactive.	Decouple authoring from generation: preview framing on existing geometry, queue jobs, stream results per chunk, prioritise the segment being edited.	Base model, fewer steps or lower resolution for drafts, Pro only for the final render; pre-generate views on a coarse grid of candidate poses.
C2 Coordinate frames (OpenCV vs. Unity left-handed Y-up) and unknown metric scale.	One conversion module with round-trip unit tests; align world-up with the gravity Puffin-World predicts; user sets floor and eye height once to fix scale.	Re-estimate poses and scale of generated frames with a feed-forward reconstructor (e.g. VGGT [6]) and register them to the scene.
C3 Rendering millions of points at 72–90 Hz per eye.	Voxel downsampling and LOD, GPU disc shader, rendering on the tethered PC.	Convert clouds to Gaussian splats rendered with the Unity plugin [7], which reports Quest 3 support.
C4 Off-distribution paths and drift (trained on 8-view sequences, one reference view); new segments may contradict geometry already shown.	Resample the spline so per-step rotation and translation stay in the training range; start each chunk from the nearest generated view; flag inconsistency by depth reprojection error.	Bound the authoring region around the seed; treat the first generation as canonical and only extend it, never regenerate.
C5 Precise framing with hand-held controllers.	Viewfinder on the controller, snapping, rule-of-thirds overlay, timeline scrubbing.	Record a path by moving head or hand in real time, then refine it.

Stretch goal: Atlas. Atlas natively takes camera poses, generates up to 1 min of 1440p video and can output point clouds or Gaussian splats [2], which would fit our editor without changing the interface. It is in early access only, and World Labs has since agreed to join AMD, so we treat access as uncertain and keep the backend model-agnostic (trajectory in, frames + depth out).

Future research. An authored camera path is one kind of action. Following our discussions with the supervisor, the same editor could later serve as an interface for studying a *unified action space* (camera, robot and vehicle actions, possibly combined, e.g. in Cosmos 3 [8]); we will keep the trajectory format extensible, but this is outside the scope of this project.

III. Outcomes and Demonstration

Expected outcome. A Unity VR application with miniature and immersive modes, keyframe authoring and asynchronous generation; a Python server around Puffin-World; a desktop baseline; and a small dataset of user-authored trajectories with their generated videos.

Technical evaluation. (i) Latency: time to first preview and time per generated chunk. (ii) Camera-control accuracy: relative pose error between the authored path and the path re-estimated from the generated frames [6], plus roll, pitch and FoV error measured with Puffin-World's own perception branch, as in [1]. (iii) Consistency: depth reprojection error between new chunks and the existing cloud.

User study. Within-subjects, about 12 participants, VR vs. desktop, counterbalanced. Tasks: reproduce a given target shot, and author a short sequence from a brief (e.g. an orbit around an object, a reveal). Measures: task time, framing accuracy (pose and FoV error to the target, overlap of the target object's bounding box), SUS [9], NASA-TLX [10] and preference. Hypothesis: VR reduces completion time for 3D framing at comparable precision.

Demo. Live at the poster session: a visitor picks one of the three scenes in the headset, places keyframes, sees the generated video replayed in VR while new geometry fills in, and exports the result. A recorded video serves as backup.

References

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